

DiD II, Portfolio Sorts & Event Study

Econometrics for Finance

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September, 2026

Identification Arsenal

Recap

Our arsenal

- Use **Matching** to create balance between treatment and control group.
- Use **Instrumental Variable (IV)** to create exogenous variation in the treatment.
- Use **Panel Data** or **Repeated Cross-section** to control for unobserved heterogeneity:
 - ▶ **Difference-in-Differences (DD)**
 - ▶ **Synthetic Controls (SC)**
 - ▶ **Event study**
- Use **Regression Discontinuity Design (RDD)** to create exogenous variation in the treatment.

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- Event Study issues

Difference-in-Difference

Estimation

Estimation:

$$y_{ist} = \alpha + \gamma_s + \lambda_t + \beta D_{ist} + \varepsilon_{ist} \quad \rightarrow \beta = \text{treatment effect}$$

where

- y_{ist} is the outcome of interest for individual i in group s at time t
- γ_s is the group fixed effect
- λ_t is the time fixed effect
- D_{ist} is the treatment dummy which is 1 for treated group after the treatment

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$$\begin{aligned} C_1 &= \alpha, & C_2 &= \alpha + \lambda, & T_1 &= \alpha + \gamma_s, & T_2 &= \alpha + \gamma_s + \lambda + \beta \\ \Rightarrow (T_2 - T_1) - (C_2 - C_1) &= \beta \end{aligned}$$

Difference-in-Difference

Another point of view (Additional concepts)

- Consider the following structure (block structure):

$$\mathbf{D} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

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- Define the CEF matrix:

$$\begin{bmatrix} \mathbb{E}[Y_{i0}|D_i = 0] & \mathbb{E}[Y_{i1}|D_i = 0] \\ \mathbb{E}[Y_{i0}|D_i = 1] & \mathbb{E}[Y_{i1}|D_i = 1] \end{bmatrix}$$

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- The Potential Outcome Matrix (POM) for $d \in \{0, 1\}$:

$$\begin{bmatrix} \mathbb{E}[Y_{i0}(d)|D_i = 0] & \mathbb{E}[Y_{i1}(d)|D_i = 0] \\ \mathbb{E}[Y_{i0}(d)|D_i = 1] & \mathbb{E}[Y_{i1}(d)|D_i = 1] \end{bmatrix}$$

Potential Outcome Matrix

Additional concepts

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- So the CEF matrix corresponds to the factual potential outcomes:

$$Y_{it} : \begin{bmatrix} \mathbb{E}[Y_{i0}(0)|D_i = 0] & \mathbb{E}[Y_{i1}(0)|D_i = 0] \\ \mathbb{E}[Y_{i0}(0)|D_i = 1] & \mathbb{E}[Y_{i1}(1)|D_i = 1] \end{bmatrix}$$

Identification

Additional concepts

- We are interested in the treatment effect:

$$\begin{aligned}\tau &\equiv \mathbb{E}[Y_{i1}(1) - Y_{i1}(0)|D_{it} = 1] \\ &\equiv \mathbb{E}[Y_{i1}(1) - Y_{i1}(0)|D_i = 1] \\ &= \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1]\end{aligned}$$

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- We can not identify τ from the CEF matrix, because we don't know the counterfactual potential outcomes.
- But we can identify τ by assuming the **parallel trends'** assumption:

$$\mathbb{E}[Y_{i1}(0) - Y_{i1}(0)|D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i1}(0)|D_i = 0]$$

Modeling counterfactual

Additional concepts

$$\begin{aligned}\mathbb{E}[Y_{i0}(0)|D_i = 1] &= \mathbb{E}[Y_{i1}(0)|D_i = 1] \\ &\quad + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|D_i = 0] \\ &= \mathbb{E}[Y_{i0}|D_i = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|D_i = 0]\end{aligned}$$

So we can identify the τ :

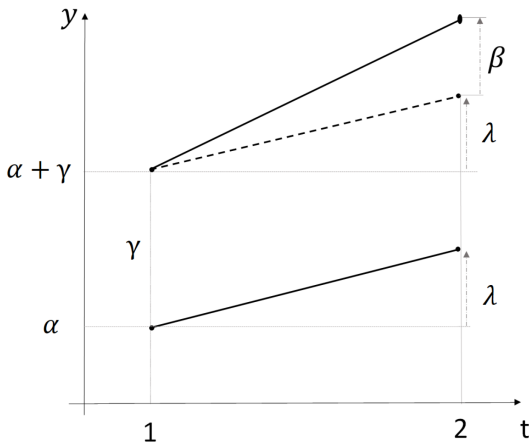
$$\begin{aligned}\tau &= \mathbb{E}[Y_{i1}(1)|D_i = 1] - \mathbb{E}[Y_{i1}(0)|D_i = 1] \\ &= \mathbb{E}[Y_{i1} - Y_{i0}|D_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0}|D_i = 0]\end{aligned}$$

So we can identify the τ by comparing the difference between the treated and control group before and after the treatment.

The main assumption

The heart of the DD is the **parallel trends** assumption which is:

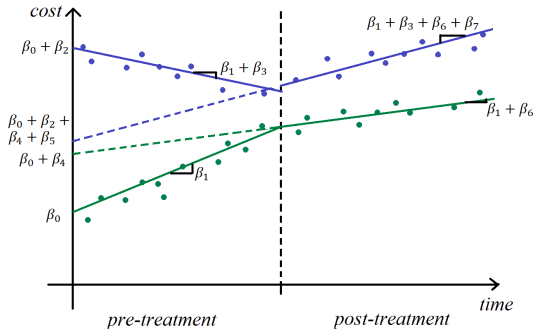
$$\mathbb{E}[y_{ist}^0|s, t] = \gamma_s + \lambda_t, \quad \mathbb{E}[y_{ist}^1|s, t] = \gamma_s + \lambda_t + \beta$$



Parallel trend assumption

- The key assumption is that the **treatment** and **control** groups would have followed the same trend in the absence of the treatment.
- This does not mean that the levels of the outcome variable are the same for the two groups.
- It is hard to test this assumption, but we can check it by plotting the outcome variable for the two groups before the treatment.
- Even if pre-trends are parallel, it is possible that other factors affect the outcome variable differently for the two groups.
- Another important assumption in DiD is exogeneity of the treatment.

Example



Additional concepts

- Dynamic Difference-in-Difference
- Triple Difference-in-Difference
- Staggered Difference-in-Difference
- Matched Difference-in-Difference
- Synthetic Control Methods

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- Another possible way to control for unobserved heterogeneity is **Regression Discontinuity Design (RDD)**.
 - ▶ RDD is a design that exploits the discontinuity in the treatment assignment
 - ▶ RDD is a non-experimental design
 - ▶ RDD is a quasi-experimental design

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 - ▶ RDD is a design that exploits the discontinuity in the treatment assignment
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RDD provides a local randomization around the cutoff point which is the cost of this method.

Basic Setup

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$$D_i = 1[R_i > c]$$

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- Examples:
 - ▶ School admission test score
 - ▶ Minimum wage
 - ▶ Scholarship awarded to students above a certain GPA

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Basic Setup

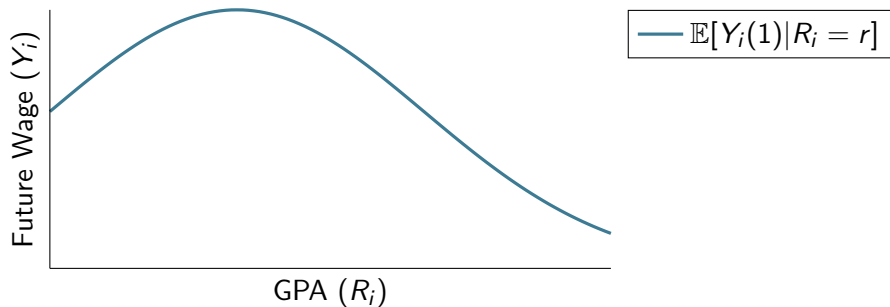
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 - ▶ School admission test score
 - ▶ Minimum wage
 - ▶ Scholarship awarded to students above a certain GPA
- We observe $Y_i(1)$ when $R_i > c$ and $Y_i(0)$ when $R_i \leq c$.
- Basic idea:
 - ▶ Compare the outcomes of individuals just above and below the cutoff point.
 - ▶ Intuitively, treatment and control groups are similar around the cutoff point (local randomization).

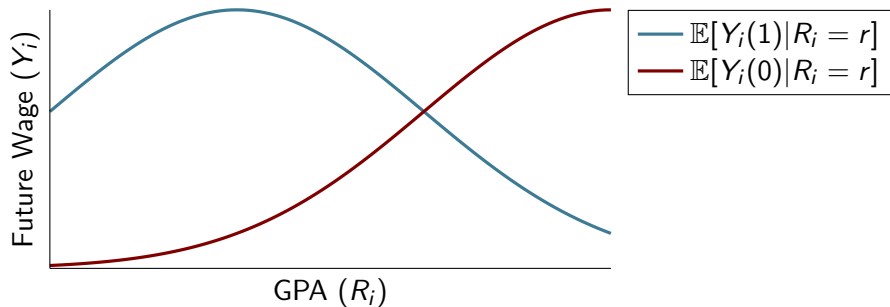
Potential Outcomes

Example

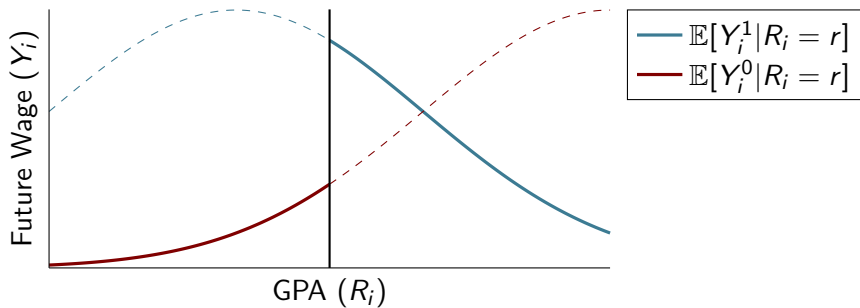


Potential Outcomes

Example



Observed Outcomes



Identification

Additional concepts

- Key assumption: continuity of the potential outcomes around the cutoff point.

$$\lim_{r \rightarrow c^+} \mathbb{E}[Y_i(d)|R_i = r] = \lim_{r \rightarrow c^-} \mathbb{E}[Y_i(d)|R_i = r], \quad d \in \{0, 1\}$$

- ▶ Potential outcomes are continuous around the cutoff point.
- ▶ The population just below must not be different from the population just above the cutoff point.

Identification

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- ▶ Potential outcomes are continuous around the cutoff point.
- ▶ The population just below must not be different from the population just above the cutoff point.
- If the continuity assumption holds, we can identify the treatment effect:

$$\begin{aligned} \lim_{r \rightarrow c^+} \mathbb{E}[Y_i|R_i = r] - \lim_{r \rightarrow c^-} \mathbb{E}[Y_i|R_i = r] \\ &= \lim_{r \rightarrow c^+} \mathbb{E}[Y_i(1)|R_i = r] - \lim_{r \rightarrow c^-} \mathbb{E}[Y_i(0)|R_i = r] \\ &= \mathbb{E}[Y_i(1)|R_i = r] - \mathbb{E}[Y_i(0)|R_i = r] \\ &= \mathbb{E}[Y_i(1) - Y_i(0)|R_i = r] = \tau \end{aligned}$$

- We need to estimate the right- and left-hand side limits

$$\lim_{r \rightarrow c^+} \mathbb{E}[Y_i | R_i = r] \quad \lim_{r \rightarrow c^-} \mathbb{E}[Y_i | R_i = r]$$

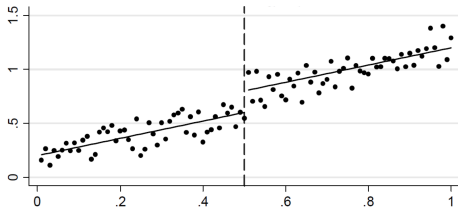
- We can estimate the limits by using the observations just above and below the cutoff point.
- We can use a linear regression model to estimate the limits:

$$Y_i = \alpha + \gamma R_i + \beta D_i + \varepsilon_i$$

- What is assumption?

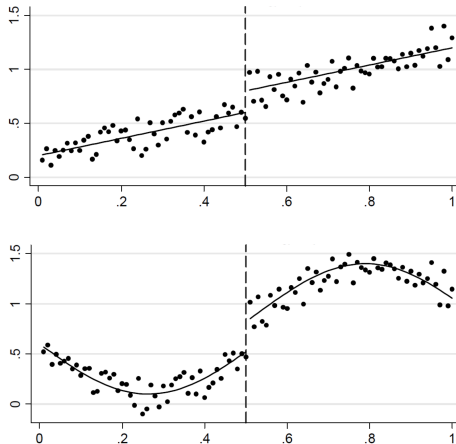
RDD

Estimation



RDD

Estimation



- In general, we can use nonlinear models to estimate the limits.

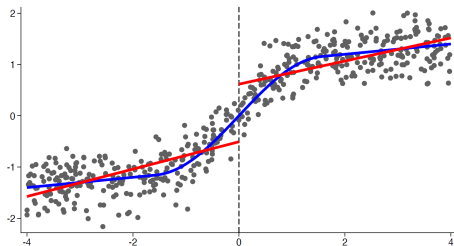
$$Y_i = f(R_i) + \beta D_i + \varepsilon_i$$

Since we extrapolate, functional form used to approximate CEF important

- Not enough to rely on OLS approximation theorems
- With insufficiently flexible specification, might mistake nonlinearity for effect
- But too flexible specification reduces precision and may be an overfitting

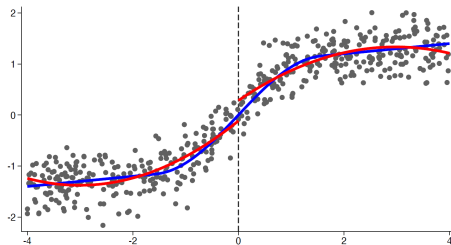
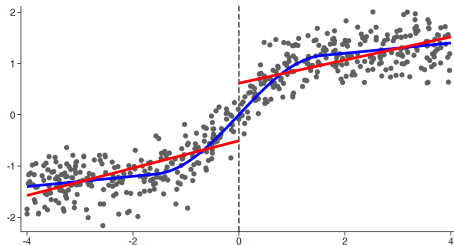
RDD

False positive



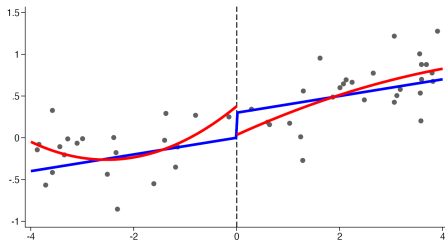
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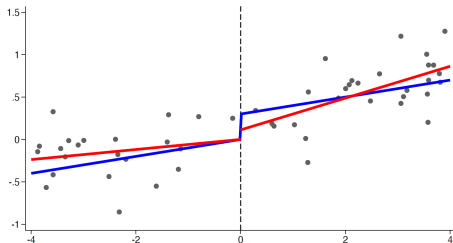
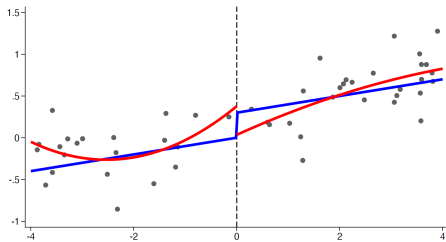
RDD

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RDD

False negative



- **Global parametric:** approximate CEF with a global parametric function
 - ▶ Traditional approach
 - ▶ Can work with moderately large datasets
 - ▶ Usually uses all data, even far from cutoff
- **Local nonparametric:** approximate CEF using nonparametric techniques
 - ▶ More recent approach
 - ▶ Requires large datasets
 - ▶ Uses only small subset of data close to the cutoff

Final point

In two last methods that we cover in this lecture, visualizing the data is very important.

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In two last methods that we cover in this lecture, visualizing the data is very important.

No one will believe you if you don't show them the data!

Assignment I

- Generate a random variable x with 1000 observations from a normal distribution with mean 0 and standard deviation 1. Also, generate a random variable $y = 2x + e$, where e is a random variable from a normal distribution with mean 0 and standard deviation 1.
- Estimate the beta coefficient of the regression of y on x .
- Now, regenerate the x with lower standard deviation and estimate the beta coefficient of the regression of y on x .
- Could you plot the average beta coefficient for different standard deviations of x ?

Assignment I

DID

- 1 This question asks you to estimate a difference in difference model. The dataset to be used is in the file "diff_in_diff_2.RData" in the folder. We have a control and treatment group observed for two time periods: one before and one after the treatment.
 - a Load the data and plot the outcome variable for the two groups before the treatment. Describe the plot.
 - b Compute the DiD estimator using the regression approach. Interpret the result.

Assignment I

Data Visualization

- ① For this question, please use the data contained in the file "industryreturns.csv". The data set contains the monthly aggregate returns (in percentage points) of 10 industries. The industries are, respectively, non-durables (1), durables (2), manufacturing (3), energy (4), hitech (5), telecoms (6), retail (7), healthcare (8), utility (9) and a category "other" for all the remaining firms (10).
 - a Convert the raw returns data in actual percentage points
 - b Examine the data. Report means by industry (e.g., 10 means). Comment on your findings.
 - c What is the volatility of the returns of the 10 industries? Comment on your findings.
 - d Standardize the returns to have mean zero and standard deviation one. Perform principal components analysis.
 - e What are the principal components' loadings?
 - f Show a scree plot of the principal components (e.g., eigenvalues on the y-axis, number of principal components on the x-axis).

Assignment II

Data Visualization

- 8 Compute the proportion of variance explained by each principal component. How many principal components do we need to explain at least 90% of the variation in the data?

Extra Assignment

Clearing the Air? The Effects of Gasoline Content Regulation on Air Quality

- ① What is the research question?
- ② What is the identification strategy? (What is the treatment and control group?)
- ③ What is your concern about the identification strategy?
- ④ What is the main result?
- ⑤ Try to replicate figure 5 and Table 2.

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Factor Models

- Factor models are a class of asset pricing models that use one or more common risk factors to explain the returns of a portfolio or a single asset.
- The idea behind the factor models is that the returns of a portfolio or a single asset are driven by the returns of the factors.
- The factors are usually portfolios of assets that are constructed based on some characteristics of the assets.
- The factor models are used to estimate the expected return of the portfolio or a single asset.

CAPM

- The Capital Asset Pricing Model (CAPM) is a factor model that uses the market portfolio as the only factor.
- The CAPM is the most famous asset pricing model and it is the foundation of modern finance.
- Assumptions:
 - ① Each investor has mean-variance preferences
 - ② All investors have a one-period investment horizon and homogeneous beliefs
 - ③ All investors are price-takers
 - ④ There are no frictions such as taxes, transaction costs, etc.
 - ⑤ The risk-free asset is in zero net supply, and the risk-free rate is exogenously given

- In the CAPM, the expected return of the asset is a linear function of the expected return of the market portfolio.

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- Where R_f is the risk-free rate, $\mathbb{E}(R_m)$ is the expected return of the market portfolio, and β_i is the sensitivity of the asset to the market portfolio.

$$\beta_i = \frac{\sigma_{im}}{\sigma_m^2} = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

CAPM

Interpretation

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Interpretation

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- The risk premium is the product of the risk premium of the market portfolio and the sensitivity of the asset to the market portfolio.
- The sensitivity of the asset to the market portfolio is the covariance of the asset with the market portfolio divided by the variance of the market portfolio.

Multi-Factor Models

- The CAPM is a single-factor model. It was the first model to use a risk factor to explain the expected return of the assets.
- The CAPM is a good model, but it has some issues.
- The basic intuition of the CAPM still carries over to multi-factor models
- Risk premium is the sum of the products of each factor's risk premium and the sensitivity of the asset to that factor.
- The sensitivity of the asset to a factor is the covariance of the asset with that factor divided by the variance of the factor. (This holds exactly when the factors are uncorrelated; otherwise the sensitivities come from a multivariate regression.)

Multi-Factor Models

General Form

$$R_i = \alpha_i + R_f + \sum_{k=1}^K \beta_{i,k} F_k + \epsilon_i$$

- R_i is the return of the asset i
- α_i is the intercept of the regression
- R_f is the risk-free rate
- F_k is the factor k and K is the number of factors
- $\beta_{i,k}$ is the sensitivity of the asset i to the **risk-factor** k

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Important Note

The factors are not firm-specific. They are common to all the firms in the market. (i.e. F_k not $F_{i,k}$)

Multi-Factor Models

- The factors are usually portfolios of assets that are constructed based on some characteristics of the assets.
- It is not necessary to use the market portfolio as one of the factors.

Examples of factors:

- Size: Small minus Big (SMB)
- Value: High minus Low (HML)
- Momentum: Winners minus Losers (WML)
- Investment: Conservative minus Aggressive (CMA)
- Profitability: Robust minus Weak (RMW)
- Term: Long-term minus short-term government bonds (TERM)
- Default: Corporate minus government bonds (DEF)
- ...

Multi-Factor Models

How to calculate the factors?

Multi-Factor Models

How to calculate the factors?

- **Sort:** Sort the assets based on the characteristics of the assets.
- **Form portfolios:** Form portfolios of assets with similar characteristics.
- **Calculate the factor:** Calculate the return of the portfolio.
- **Repeat:** Repeat the process for different characteristics.

More details at the following link: [*Construction of the Fama-French-Carhart four factors model for the Swedish Stock Market using the Finbas data*](#)

You can download the factors for the US market from [Ken French's website](#).

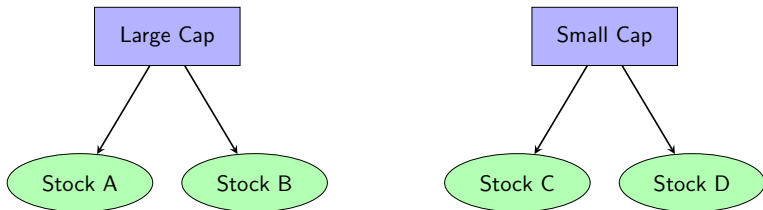
Factor construction

Size Example



Factor construction

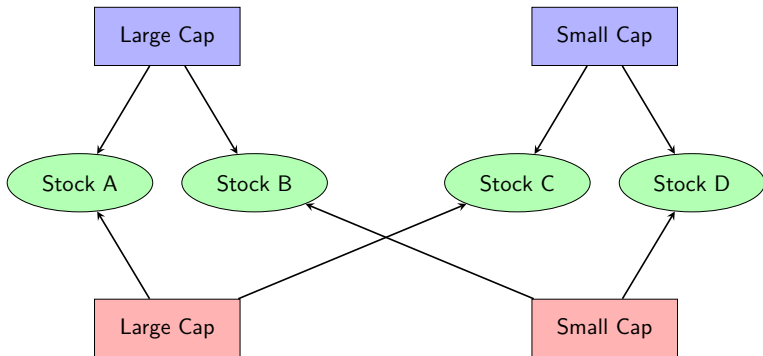
Size Example



- Blue rectangles are portfolios that are constructed based on the size of the firms at a fixed point in time.

Factor construction

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- Red rectangles are portfolios that are constructed based on the size of the firms at a later point in time.

Table of Contents

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- Estimation
- Hypothesis Testing
- Cross-Sectional Regression
- Event Study issues

Event Study

- An event study is a statistical method to assess the impact of an event on the value of a firm.
- The event is usually a corporate event such as an announcement of earnings, a merger, a new product, or a change in management. But it can be any event that is likely to have an impact on the value of the firm and is unanticipated by the market (e.g. a law change).
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Main assumption

Market is efficient and the price of the stock reflects all the available information.

Notation and definitions

- **Event:** The event should be a well-defined and well-publicized event that is likely to have an impact on the value of the firm, and the event date should be precisely defined.
 - ▶ e.g. Earnings announcement, dividend announcement, merger announcement, etc.
 - ▶ We should define the event window, which is the period of time around the event date. (e.g. 5 days before and 5 days after the event date)
 - ▶ The length of the event window will be determined by the nature of the event.

Notation and definitions

- **Return:** The return is the percentage change in the firm's stock price, including dividends, over the event window. (R_{it})
 - ▶ We can compute the return for each day in the event window or the average over the event window.
 - ▶ e.g. we might want to investigate the return for all the five days before the event up to five days after the event date.
($t = -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$)

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 - ▶ In most cases, we need to be able to separate the return of the firm from the return of the market.
 - ★ For example, if following the event date, the firm's stock price increases by 2% and the market increases by 2%, then it would be misleading to conclude that the event led to an increase in the firm's stock price.

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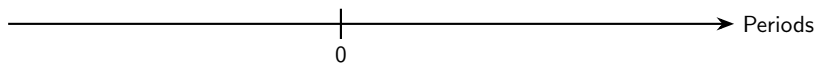
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- There is typically a gap between the estimation window and the event window to avoid the contamination of the estimation window by the event.
- Clearly, what we would like to do is to calculate the return that would have been expected for that stock if the event had not happened at all, so that we can isolate the impact of the event from any unrelated incidents that may be occurring at the same time.

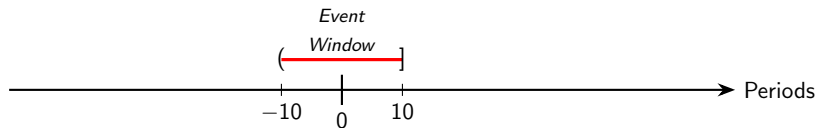
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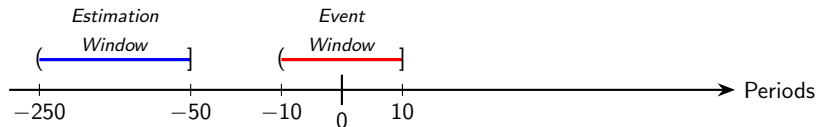
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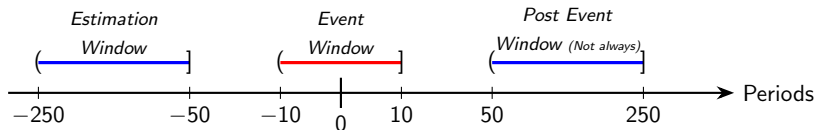
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- **Control Portfolios:** Use a portfolio of firms that are similar to the firm that we are investigating to estimate the expected return.

Hypothesis Testing

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- The test statistic is:

$$\widehat{SAR}_{it} = \frac{AR_{it}}{\hat{\sigma}_{AR_{it}}} \sim N(0, 1)$$

Variance of the abnormal return

- The variance of the abnormal return is:

$$\hat{\sigma}_{u_i}^2 = \frac{1}{T-2} \sum_{t=1}^T \hat{u}_{it}^2$$

where T is the number of observations in the estimation period and $\hat{u}_{it}$ are the market-model residuals.

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- An adjustment is made to the variance to reflect the errors arising from the estimation of α and β in the market model ($\hat{\sigma}_m^2$ is the variance of the market return over the estimation window):

$$\hat{\sigma}_{AR_{it}}^2 = \hat{\sigma}_{u_i}^2 \left[1 + \frac{1}{T} + \frac{(R_{mt} - \bar{R}_m)^2}{T \hat{\sigma}_m^2} \right]$$

Cumulative Abnormal Return

- It is likely that there will be quite a bit of variation of the returns across the days within the event window, with the price rising on some days and then falling on others.

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- We may therefore consider computing the cumulative abnormal return over a multi-period event window by summing the abnormal returns over several periods, say from time T_1 to T_2 .

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- It is common to examine
 - ▶ a pre-event window (to consider whether there is any anticipation of the event) and a post-event window
 - ▶ i.e. we sum the daily abnormal returns for a given firm i for days $t - 10$ to $t - 1$, say, and separately from $t + 1$ to $t + 10$, with the actual day of the event, t , being considered on its own.

Average Abnormal Return

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- **Average Cumulative Abnormal Return:** The average cumulative abnormal return is the average of the cumulative abnormal return of all the firms in the sample.

$$\begin{aligned} \widehat{CAR}(T_1, T_2) &= \frac{1}{N} \sum_{i=1}^N \widehat{CAR}_i(T_1, T_2) \\ \hat{\sigma}_{CAR(T_1, T_2)}^2 &= \frac{1}{N^2} \sum_{i=1}^N \hat{\sigma}_{CAR_i(T_1, T_2)}^2 \\ \Rightarrow \widehat{SCAR}(T_1, T_2) &= \frac{\widehat{CAR}(T_1, T_2)}{\hat{\sigma}_{CAR(T_1, T_2)}} \sim N(0, 1) \end{aligned}$$

Cross-Sectional Regression

- We learned how to estimate the expected return and the abnormal return for a single firm. But what if the event has heterogeneous effects across firms?

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- **Cross-Sectional Regression:** The cross-sectional regression is a regression of the abnormal return of the firm on the characteristics of the firm.

$$AR_i = \alpha + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \epsilon_i$$

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- We could use cumulative abnormal return instead of abnormal return.
- It is recommended to use heteroskedasticity-robust standard errors in the regression.

Event Study issues

Cross-Sectional Dependence

The cross-sectional regression is a common method to investigate the impact of the event on multiple firms. But there are some issues with this method.

- **Cross-Sectional Dependence:** The abnormal returns of the firms are correlated with each other. This correlation could be due to common market-wide or industry-wide shocks, and it is especially severe when the events cluster in calendar time.
 - ▶ One solution is to not use it!
 - ▶ Construct portfolios of firms having the event at the same time, and then perform the analysis on each portfolio

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- I have read papers with around 200 observations that use the standard tests for hypothesis testing. But they are old!
- There are suggestions to use the bootstrap method to estimate the distribution of the test statistics or to use non-parametric tests.

Further issues

- **Variance of Returns:** The event itself raises the volatility of returns, so the variance estimated over the estimation window understates the true variance during the event window. The measured variance will be too low and the null hypothesis of no abnormal return during the event will be rejected too often.

Long Event windows

- For short periods, model discrepancies are minor. However, for longer durations, small model errors can significantly affect the calculation of abnormal returns and the event's perceived impact. Proper model design and risk allowance are crucial for long-term assessments.

Long Event windows

- For short periods, model discrepancies are minor. However, for longer durations, small model errors can significantly affect the calculation of abnormal returns and the event's perceived impact. Proper model design and risk allowance are crucial for long-term assessments.
- Two approaches to long event windows:
 - ① **Cumulative Abnormal Returns (CARs)**
 - ② **Buy-and-Hold Abnormal Returns (BHARs)**

$$\widehat{BHAR}_i(T_1, T_2) = \left[\prod_{t=T_1}^{T_2} (1 + R_{it}) - 1 \right] - \left[\prod_{t=T_1}^{T_2} (1 + \mathbb{E}[R_{it}]) - 1 \right]$$

- ★ When constructing BHARs the expected return is based on a non-event firm or portfolio of firms that is matched in some way to the event firm
- ★ Or you can create a portfolio of event firms and use the models that we talked about to estimate the expected return.

Event study implementation

- There are tools such as [Event Study Tools](#) for running an event study.
- But as you can see from the lecture, you as a researcher should decide about many things in the event study.
- So you should be aware of the problems and the assumptions behind the event study when you use such tools.

Assignment I

- ① Investigate the effect of Brexit on U.K. banks
 - ▶ Calculate the abnormal return for the U.K. banks after the Brexit referendum
 - ▶ Calculate the abnormal return for the U.K. banks after the first U.K. general election following the Brexit referendum
 - ▶ Calculate the abnormal return for the U.K. banks after Boris Johnson resigned as foreign secretary
 - ▶ Calculate the abnormal return for the U.K. banks after the publication of the Brexit withdrawal agreement
 - ▶ What is your conclusion?

Extra Assignment

- ① Let's calculate the abnormal return for the firm that filed a patent. We will use the WRDS to implement the event study.
 - a Download `firmEvent.csv` from Canvas. It contains the firm identifier (`permno`) and the event date (`date`).
 - b Go to [WRDS event study page](#) and answer the main questions.

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